

A Strictly Sofic Mod-Three Gap Suspension and Its Exact Cover Correction

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Abstract

We construct an exact nonlattice suspension over the binary mod-three gap shift, a strictly sofic source. Its minimal three-state residue cover has determinant $1 - u - v^3$, but the cover has three lifts of the all-zero label fixed point. Correcting that one exceptional orbit at every period gives the intrinsic rational zeta $(1 + v + v^2)/(1 - u - v^3)$ and an exact primitive label-orbit product. We keep the cover determinant, intrinsic inverse zeta, and any hypothetical natural Fredholm owner distinct. No target or arithmetic factorization is claimed.

1 A strictly sofic source and its cover

Let $X_3 \subset \{0, 1\}^{\mathbb{Z}}$ contain the sequences whose finite zero runs between consecutive ones have length in $3\mathbb{Z}_{\geq 0}$; include the all-zero point. It is presented by residue states 0, 1, 2 and labeled edges

$$0 \xrightarrow{1} 0, \quad 0 \xrightarrow{0} 1, \quad 1 \xrightarrow{0} 2, \quad 2 \xrightarrow{0} 0. \quad (1)$$

The graph is right resolving. Its follower sets are distinct: the shortest words ending at the next 1 from states 0, 1, 2 are respectively 1, 001, 01. Hence it is the minimal follower-separated right-resolving cover.

The shift is not of finite type. For any proposed memory L , choose a periodic zero gap $3m + 1 > 2L$. The gap violates the mod-three rule, but every length- $(L + 1)$ block contains at most one 1 and embeds in an allowed multiple-of-three gap (or in the all-zero point). No finite local rule can therefore define X_3 .

Let u mark label 1 and v label 0. The cover matrix is

$$B(u, v) = \begin{pmatrix} u & v & 0 \\ 0 & 0 & v \\ v & 0 & 0 \end{pmatrix}, \quad D_{\text{cov}}(u, v) = \det(I - B) = 1 - u - v^3. \quad (2)$$

2 The exceptional all-zero orbit

Any label sequence containing a 1 has a unique cover lift: that coordinate forces state 0, and the zero residues propagate in both directions. The all-zero label sequence is different. It is one label fixed point, but has three cover phase lifts forming a single cover orbit of least period three.

Let $F_n(u, v)$ be the intrinsic weighted sum over label points fixed by the n th shift power. The ordinary points agree with their unique cover lifts. The all-zero label point contributes v^n for every n , while its three cover lifts contribute $3v^n$ exactly when $3 \mid n$. Therefore, for every $n \geq 1$,

$$F_n(u, v) = \text{Tr } B(u, v)^n + (1 - 3\mathbf{1}_{\{3 \mid n\}})v^n. \quad (3)$$

This classification proves (3) without a finite-period inference.

3 Intrinsic zeta and primitive product

Define $\log Z_{140} = \sum_{n \geq 1} F_n/n$ in the total-label-degree completion. The cover part is $-\log(1 - u - v^3)$. The exceptional contribution is

$$\sum_{n \geq 1} \frac{v^n}{n} - \sum_{k \geq 1} \frac{3v^{3k}}{3k} = -\log(1 - v) + \log(1 - v^3) = \log(1 + v + v^2).$$

Hence

$$Z_{140}(u, v) = \frac{1 + v + v^2}{1 - u - v^3}, \quad D_{140} = Z_{140}^{-1} = D_{\text{cov}} \frac{1 - v}{1 - v^3}. \quad (4)$$

Every periodic label point has a unique primitive label orbit and repetition exponent. Grouping the fixed-point logarithm by that root gives the exact formal product

$$D_{140}(u, v) = \prod_{[\gamma] \text{ primitive label}} \left(1 - u^{N_1(\gamma)} v^{N_0(\gamma)}\right). \quad (5)$$

The regrouping is coefficientwise finite. Equation (4) does not assert that D_{140} is the determinant of a separately constructed natural Fredholm operator on intrinsic label space.

4 Suspension, validation, and boundary

Assign roof times $\tau(1) = 1$ and $\tau(0) = \sqrt{2}$, and substitute $u = ze^{-s}$, $v = ze^{-\sqrt{2}s}$. Equation (5) becomes

$$D_{140}(z, s) = \prod_{[\gamma]} \left(1 - z^{|\gamma|} e^{-s(N_1 + \sqrt{2}N_0)}\right).$$

The label fixed cycles $[1]$ and $[0]$ have lengths 1 and $\sqrt{2}$, proving the roof nonlattice. The positive roof entropy parameter is the unique root of $1 - e^{-h} - e^{-3\sqrt{2}h} = 0$ because the left side rises continuously from -1 to 1 .

Exact replay through period fifteen contains 969 intrinsic rooted points, 74 primitive label cycles, 60 rooted label-count cells, and 32 primitive cells. The independent checker passes 2,028 assertions, SymPy passes 53 checks, byte replay passes, and all 54 hostile cases (53 repaired-hash and one stale-hash) are rejected. The prefix tests implementation only; (3)–(5) hold at every period by proof.

The strict tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), overall ROUTE_A_EXPLORATORY. We claim no target divisor, functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, natural self-adjoint or unitary lift, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.